

Synthetic Data Generation and Vortex Initialization



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Introduction:

- In recent decades, track prediction has been steadily improved. However, there has been little improvement in predicting the intensity of cyclones.
- Primary reason is that the intensity depends on the inner-core dynamics, whereas the track prediction depends more on the large-scale environments.
- The inner-core dynamics are mainly associated with vortex structure, intensity and vortex size.
- It has been suggested that the accurate representation of the inner-core structure of tropical cyclones in the initial conditions is as important as the representation of the large-scale environment.
- To improve the representation of the initial vortex, a bogus vortex is often applied because the initial vortices provided by the large-scale analysis are often too weak or misplaced.
- The advantages of the use of bogus vortices are the ability to customize the vortex structure and low computational cost.

TC Vortex Initialization

- The synthetic vortex should possess three properties to minimize dynamic adjustment and false spin-up ([Kurihara et al. 1993](#)) :
 - Structural consistency,
 - Resemblance to the “real storm,”
 - New vortex should be compatible with the numerical model.
- To ensure these qualities are enforced in the generation of TC flows for model initialization, three general techniques, that work together or alone, have been developed: **Data Assimilation, Vortex Bogusing and Dynamic Initialization.**

Bogus Vortex

- **Bogus vortex is constructed by inserting an analytical vortex structure into the global/regional analysis fields at the initial forecast time like Holland and Rankine vortex schemes.**
- **Simple assumed hydrostatic, gradient wind, and thermal wind balances; TC intensity matched the observed.**
- **Initial adjustments with model dynamics and physics; arbitrary horizontal/vertical vortex structure, without the use of any information of the environment the TC is embedded.**

Holland Vortex

- The Holland vortex (Holland, 1980) gives the surface pressure profile $P_s(r)$ as a function of the radius.

$$P_s(r) = P_c + (P_n - P_c)\exp(-a/r^b)$$

P_n : Ambient pressure (pressure of the outermost closed isobar),

a and b : scaling parameters and

P_c : Central pressure.

- The wind profile is generated from the gradient wind relationship and is given as

$$V = \left[ab(P_n - P_c)\exp(-a/r^b) / \rho r^b + r^2 f^2 / 4 \right]^{1/2} - rf / 2$$

f : Coriolis parameter and ρ : Density of air.

- In the region of maximum wind, f can be neglected as the pressure gradient force is balanced by the centrifugal forces and the vortex is assumed to be in cyclostrophic balance.

- Thus the surface winds are given by

$$V = [ab(P_n - P_c)\exp(-a/r^b)/\rho r^b]^{1/2}$$

- The parameters 'a' and 'b' are related to radius of maximum wind (RMW) in a cyclone by the following relation. 'b' value affects the profile shape and is a function of the cyclone intensity.

$$\text{RMW} = (a)^{1/b}$$

- The parameter 'b' usually lies between 1.0 and 2.5 (*Agnihotri et al. 2008*).
- These parameters have to be determined empirically and are different for different ocean basins and also vary for each cyclone

Rankine vortex

- The Rankine (Depperman 1947 & DeMaria 1985) vortices generate the pressure observations from the wind relationship.
- The wind profile is given as

$$V = V_m \left(\frac{r}{r_m} \right) \quad r < r_m$$
$$V = V_m \left(\frac{r}{r_m} \right)^{-\alpha} \quad r \geq r_m$$

V_m : Max. tangential velocity; r_m : Radius of Max. wind; α : decaying factor, which is to be determined empirically from the observations and usually lies between 0.3 and 0.8 (*Gray and Shea 1973*). r : Distance from the TC center.

- The vortex component derived from the symmetric Rankine vortex is controlled by maximum tangential velocity.

- It is estimated based on the following empirical formula developed by Mishra and Gupta (1976) for the Indian seas.

$$V_m = 14.2 (P_n - P_c)^{1/2}$$

- The surface pressure P_s is calculated from the gradient wind relation. In the inner region ($r < r_m$) the surface pressure is given as

$$P_s(r) = \rho (V_m + f V_m) \left(\frac{r}{r_m} \right)^2 + P_c$$

- Similarly in the outer region ($r \geq r_m$), the surface pressure is given by

$$P_s(r) = \rho f r_m \left(\frac{r}{r_m} \right)^{1-\alpha} - \rho \left(\frac{r}{r_m} \right) - 2^\alpha + B$$

Where

$$B = P_c + \frac{\rho}{2} V_m (V_m + f r_m) - \rho f \frac{V_m r_m}{(1 - \alpha)} + V_m^2 \frac{\rho}{2\alpha} \quad \text{and at } r = 0, \quad P_s = P_c$$

- f is the Coriolis parameter at the latitude of the storm centre and ρ is the density of air. The wind and the pressure profiles are expressed as a function of P_c , V_m , r_m and α .

Dynamical vortex initialization (DVI)

- Axisymmetric vortex is spun up with the axisymmetric version of the 3D forecast model or a different 3D model (Kurihara et al. 1993 or Hendricks et al. 2011).
- More realistic 3D TC structure and dynamical balance but initial adjustment issue; no information of the environment (asymmetric structure) the TC is embedded; cold model start up.
- Therefore, 6 hrs (t_0-6) cycle with a warm start model forecast initialization applied to produce:
 - Dynamical balance (after 6 –h spin-up)
 - Asymmetric structure forced by the environment
 - Warm model physics startup (after 6-h integration)

Variational Bogus Data Assimilation (BDA)

- BDA uses variational data assimilation with synthetic observations of a TC vortex that closely matches the observed TC intensity and structure.
- The TC is dynamically balanced and more consistent with model physics.
- Variational data assimilation generally uses simplified (linearized) model physics which are quite nonlinear.
- Observations in TC core are often rare and the synthetic data are just better than nothing since few observations.

TC Vortex Initialization Scheme at NCMRWF

- The insertion of a realistic synthetic vortex into the numerical model consists of the following 4-steps:
 - Remove spurious vortices smoothly from the analysis but retain the surrounding larger-scale fields (i.e environmental fields) intact.
 - A three-dimensional axisymmetric vortex is constructed from available operational estimates of vortex size and strength with the help of observations.
 - Relocation of inner core to observed position (vortex center matches the vortex drift speed).
 - The artificial vortex fields are blended smoothly into the corresponding environmental fields and handed to the model initialization procedure.

- **TC Vortex Initialization scheme input from NCUM analysis:**
 - U, V, T, Geop. Height and RH at all pressure levels.
 - Surface Temp. , Surface Pressure and MSLP
- **Input from observations (Tcvital file):**
 - Latitude & Longitude (Int. time, -6hrs & -12hrs)
 - Central pressure; Environmental pressure (hPa)
 - Radius of outermost closed isobar (ROCI; km)
 - Estimated Maximum wind speed (MSW; m/s)
 - Radius of Max. wind (km)
- **Vortex Initialization (VI) Method of Davidson et al. (MWR, 2014) is used to create the synthetic vortex in NCUM.**
- **The vortex structure is generated based on observed location, central pressure and radius of outer closed isobar.**
- **Synthetic observations of Mean Sea Level Pressure (MSLP) produced by VI scheme for the cyclone were used in 4D-Var data assimilation system of both Global (25 km) & Regional (12 km) models.**

- **The vortex relocation procedure is performed only when:**

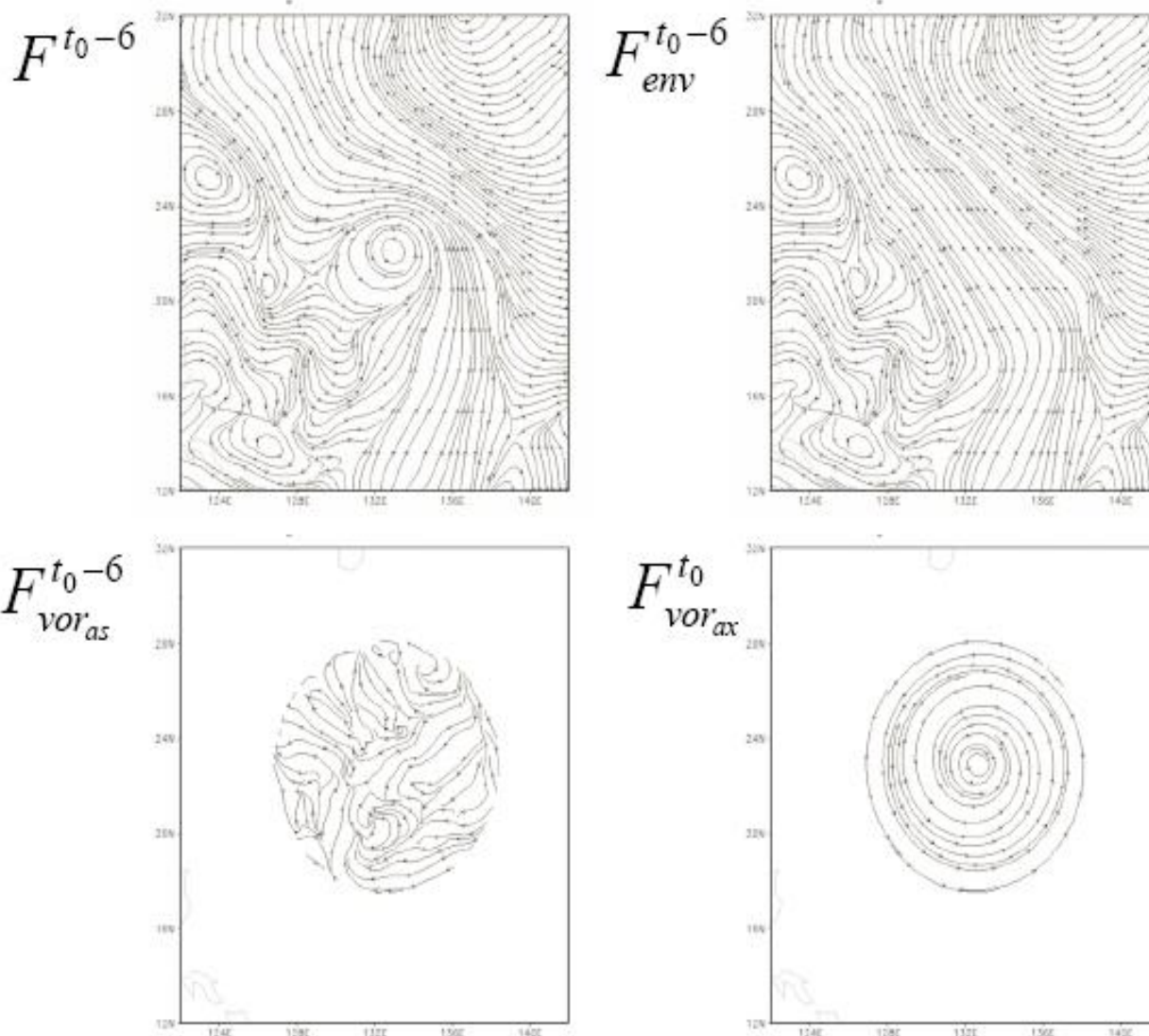
- The first-guess TC center is more than one grid distance away from the observed location [(Obs – first-guess) > 1 grid at least].
- Maximum wind speed of the first-guess TC at 850 hPa should be greater than 15 m/s.
- TC center is at least 300 km away from the lateral boundary.
- There are no grid points with terrain height greater than 50 m within 150 km of the TC center.
- The TC Centre is not too close to the land (< 200 km).

- **The high terrains near a TC center may affect the vortex relocation in three ways.**

- ❖ The low-level wind distribution is greatly modified by the high terrains that the TC edge cannot be properly determined.
- ❖ TC edge determined at 850 hPa may not be applicable to all levels.
- ❖ Large gradient of pressure, height and temperature on model levels at a high terrain point that creates imbalances in the relocated flow.

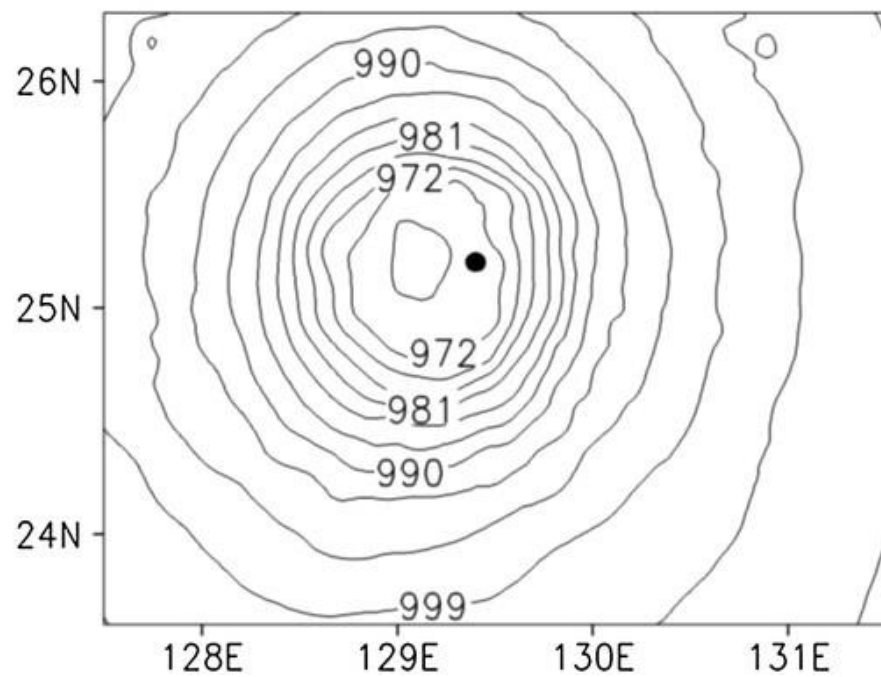
Initial field separation

500 hPa Streamline

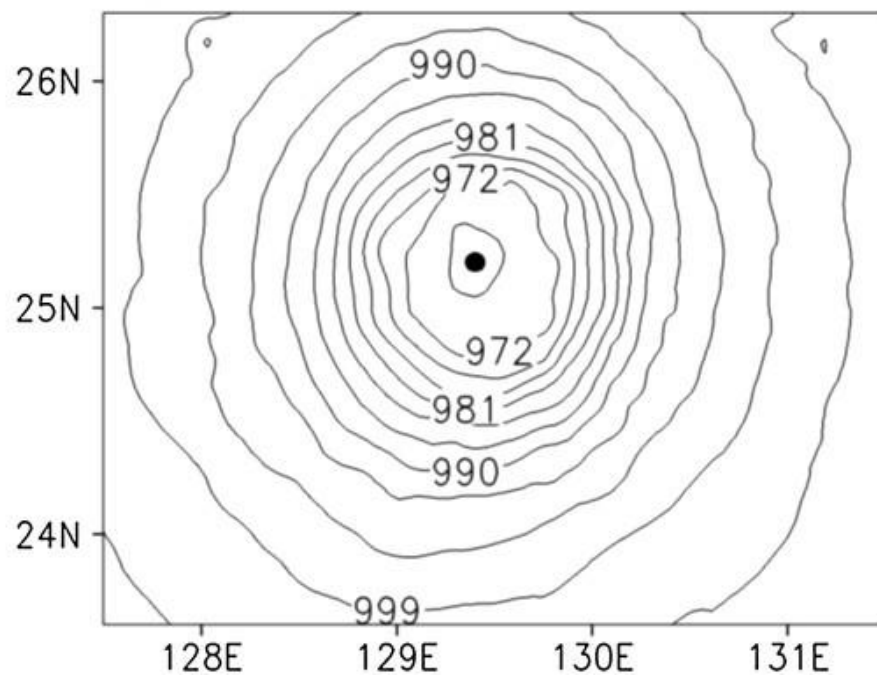


$$F^{t_0-6} = F_{env}^{t_0-6} + F_{vor_{as}}^{t_0-6} + F_{vor_{ax}}^{t_0}$$

a) Before Relocation



b) After Relocation

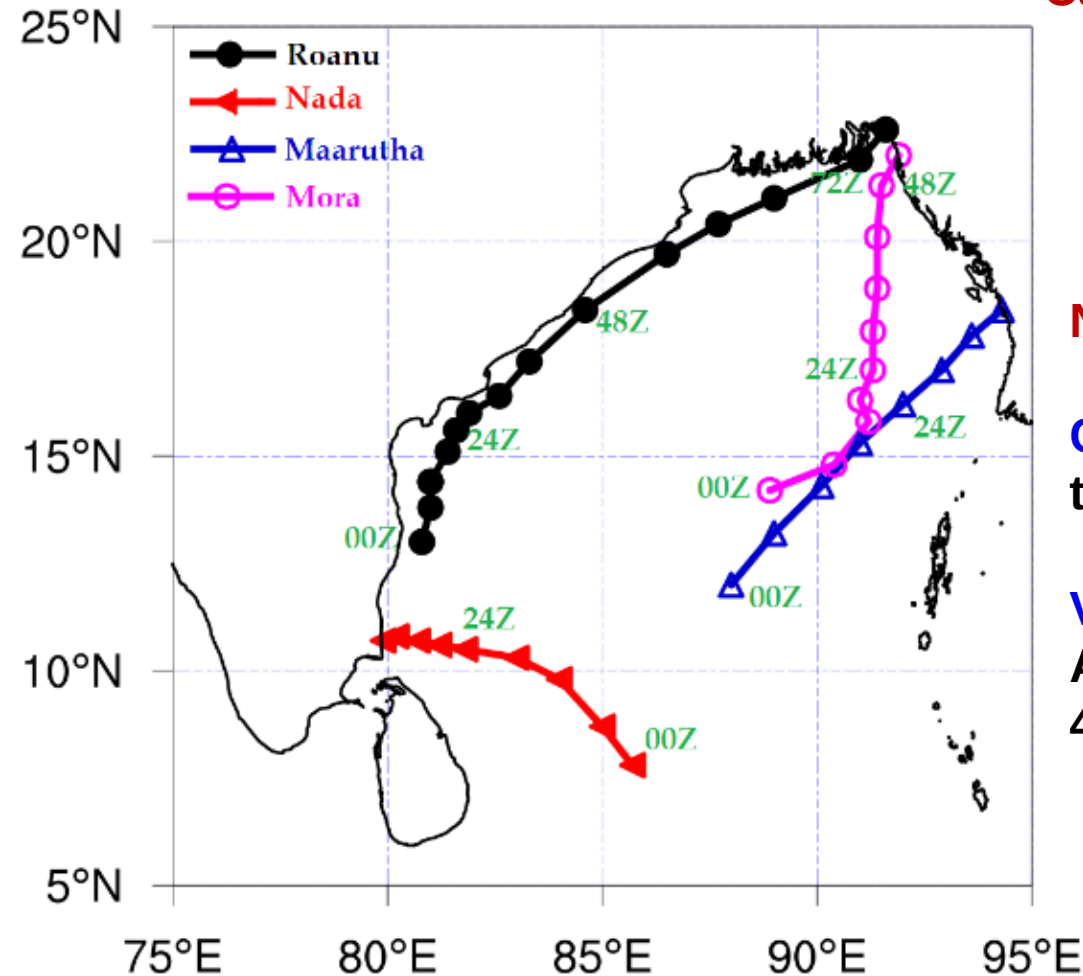


Impact of Vortex Initialization (VI) in Prediction of TCs over BoB with NCUM Model

Details of the model simulations and observed landfall time of BoB TC cases

Cas e	TC name	Duration of TC	Simulation period in 24-h intervals, (No. of Initial Conditions)	Observed Landfall Place name, Lat-Lon and timings
1.	Roanu	17-22 May, 2016	19-21May, 2016 (3)	Southeast coast of Bangladesh (22.60N & 91.6E) around 1000 UTC of 21st May, 2016
2.	Nada	29Nov-02Dec, 2016	30 Nov-01Dec, 2016 (2)	North Tamil Nadu coast near Nagapattinam (10.75N & 79.9E) between 2230 and 2330 UTC of 01st Nov, 2016
3.	Maarutha	15-17 April, 2017	16 April, 2017 (01)	Myanmar coast near Sandoway (Thandwe, 18.4⁰N & 94.3⁰E) between 1800-1900 UTC of 16th Apr, 2017
4.	Mora	27-31 May, 2017	28-29 May, 2017 (2)	South of Chittagong, Bangladesh in the forenoon (22.0⁰N & 91.9⁰E) between 0400 and 0500 hrs UTC) of 30th May, 2017.

Observed Track (IMD)



Case-1: Roanu (17-22 May 2016)

Case-2: Nada (29 Nov. - 02 Dec. 2016)

Case-3: Maarutha (15 - 17 April 2017)

Case-4: Mora (28-31 May, 2017)

Numerical Expts.

CNTL: Assimilated GTS Obs. through 4DVAR.

VOTX: Applied Bogusing plus Assimilation of GTS Obs. through 4DVAR.

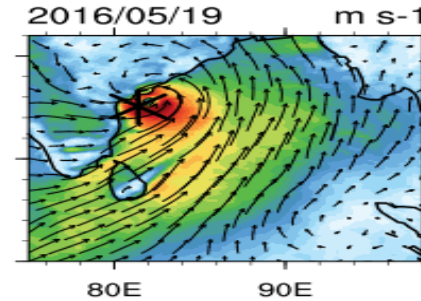
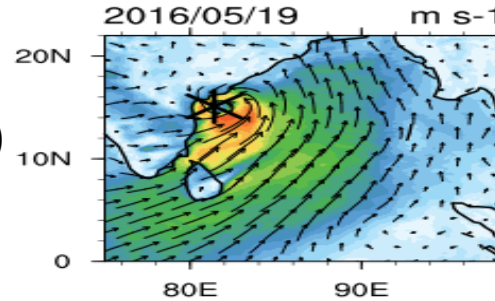
10 m surface wind at model initial time

CNTL

VOTX

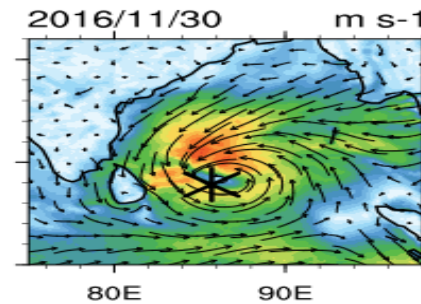
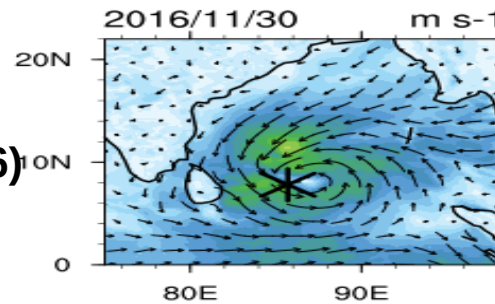
Case-1

(00 UTC 19 May 2016)



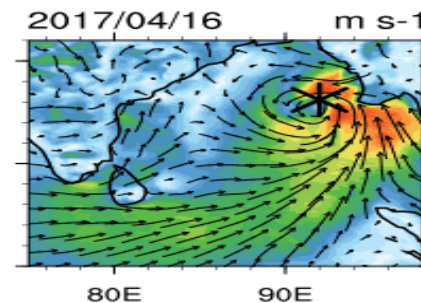
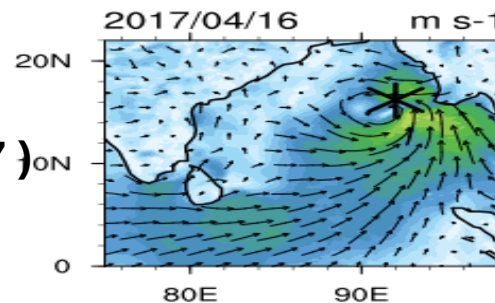
Case-2

(00 UTC 30 Nov. 2016)



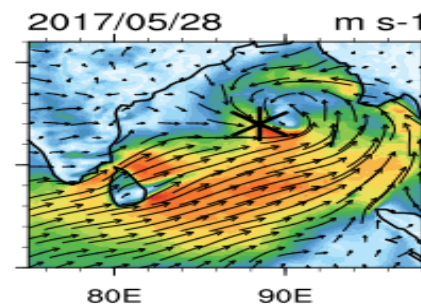
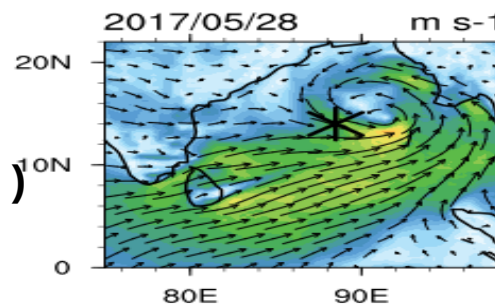
Case-3

(00 UTC 16 April 2017)

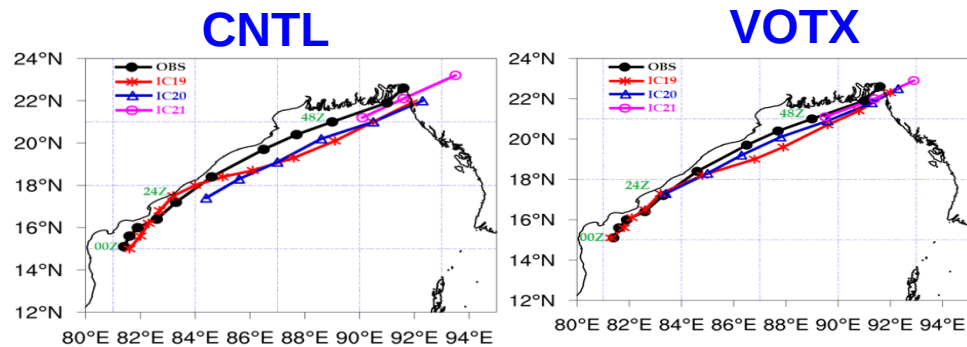


Case-4

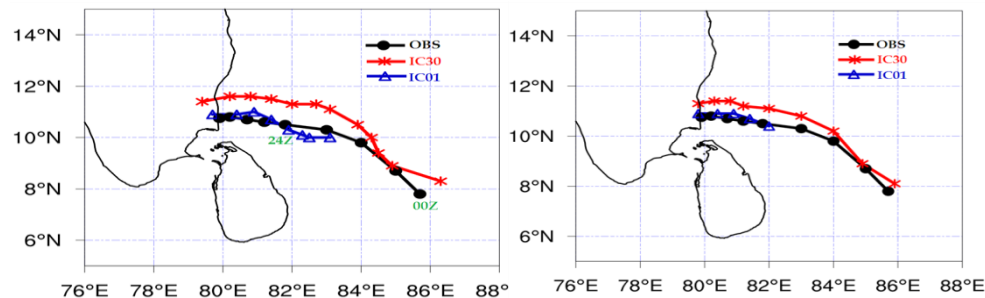
(00 UTC 28 May 2017)



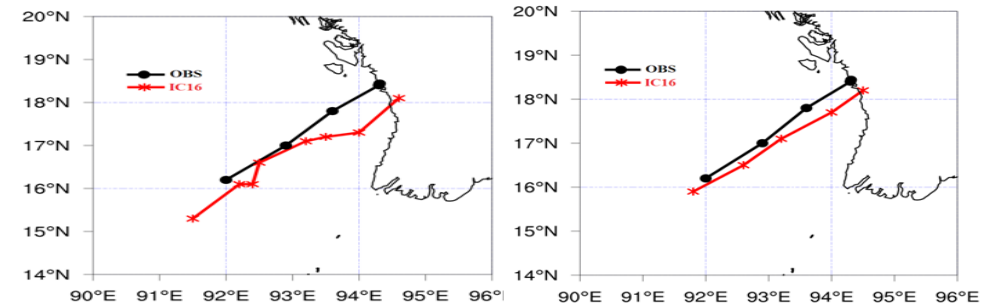
Case 1



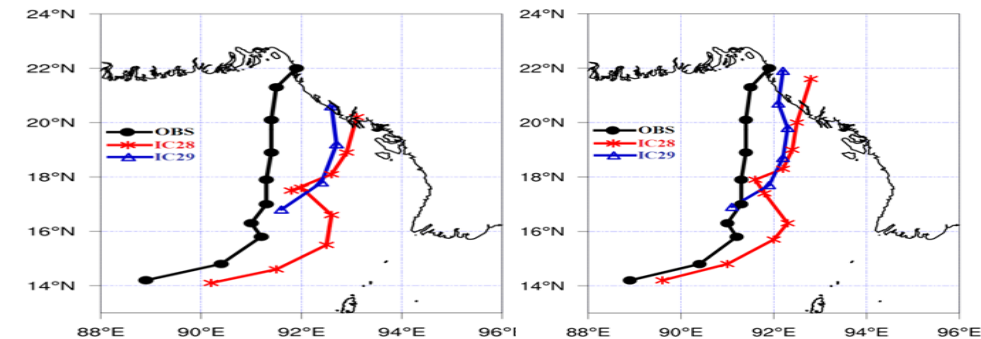
Case 2

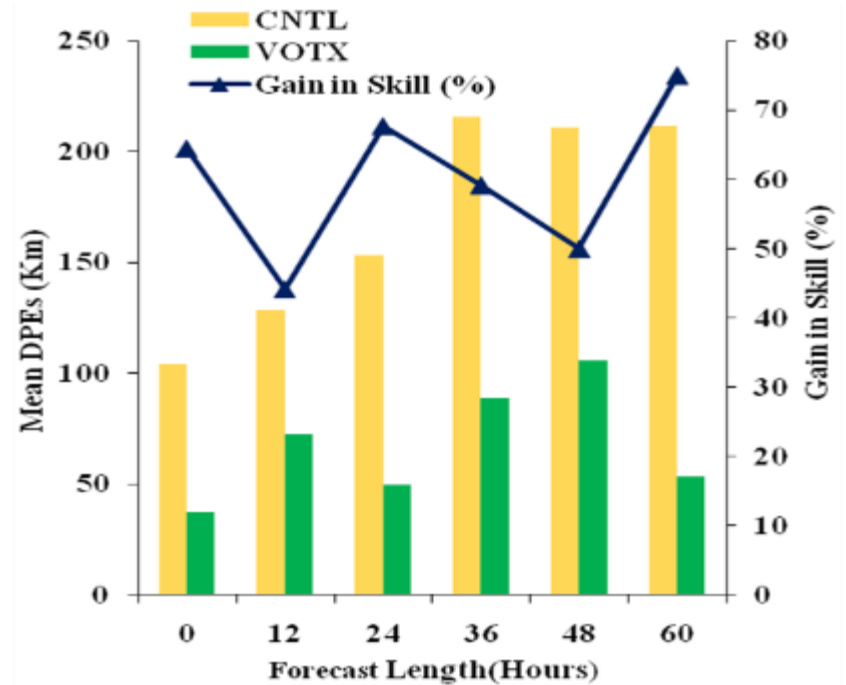
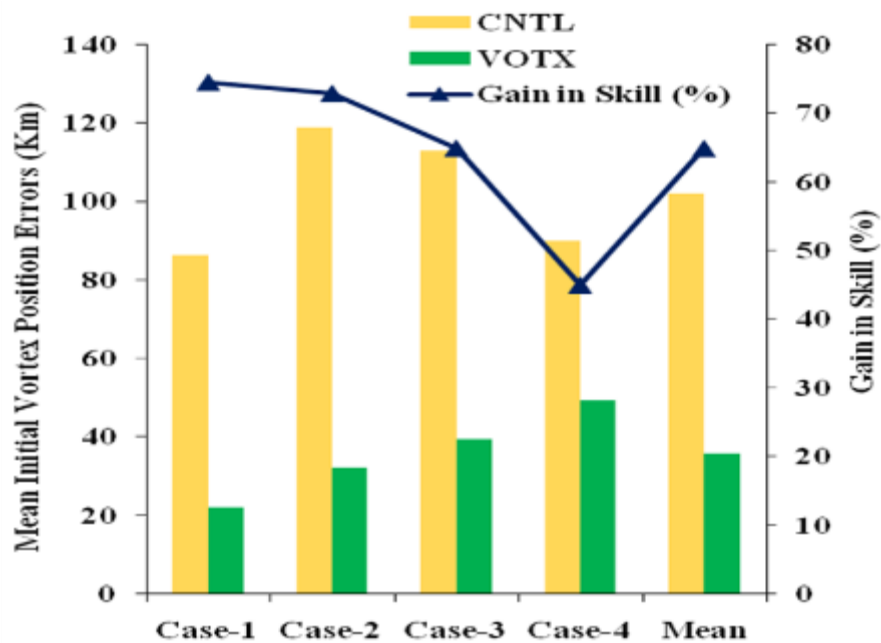


Case 3

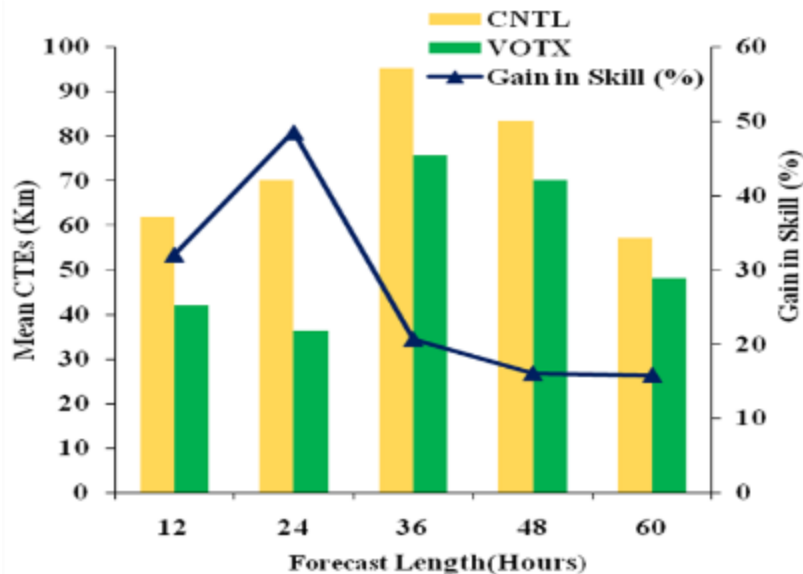


Case 4

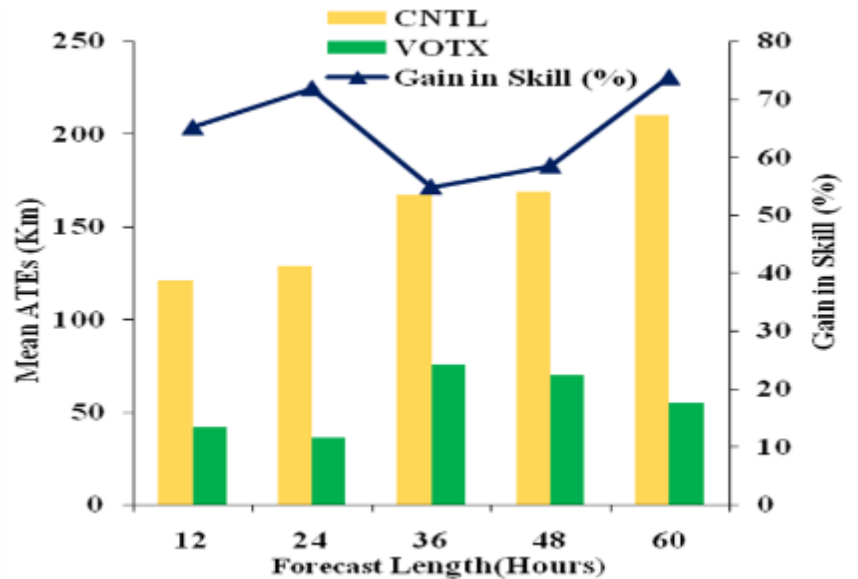


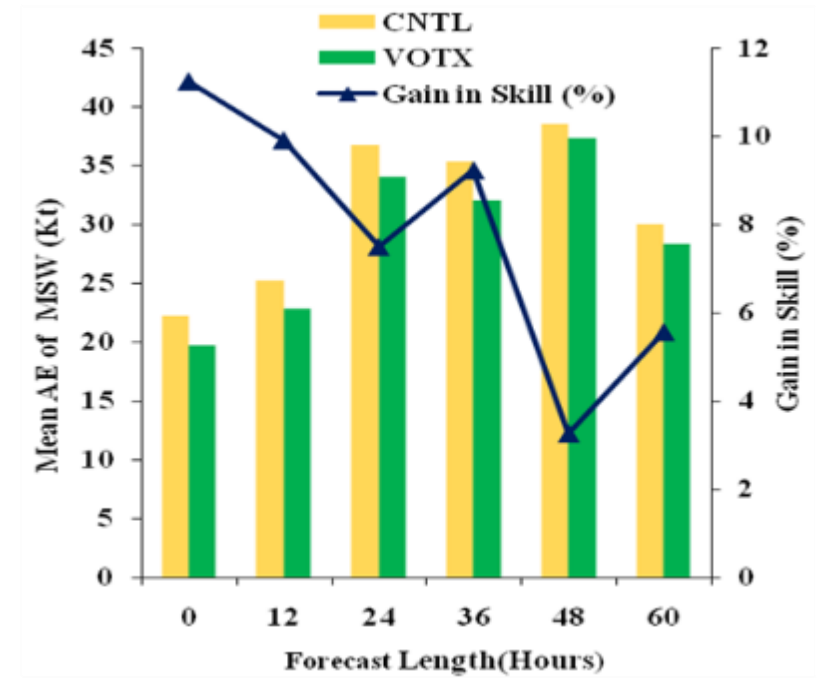
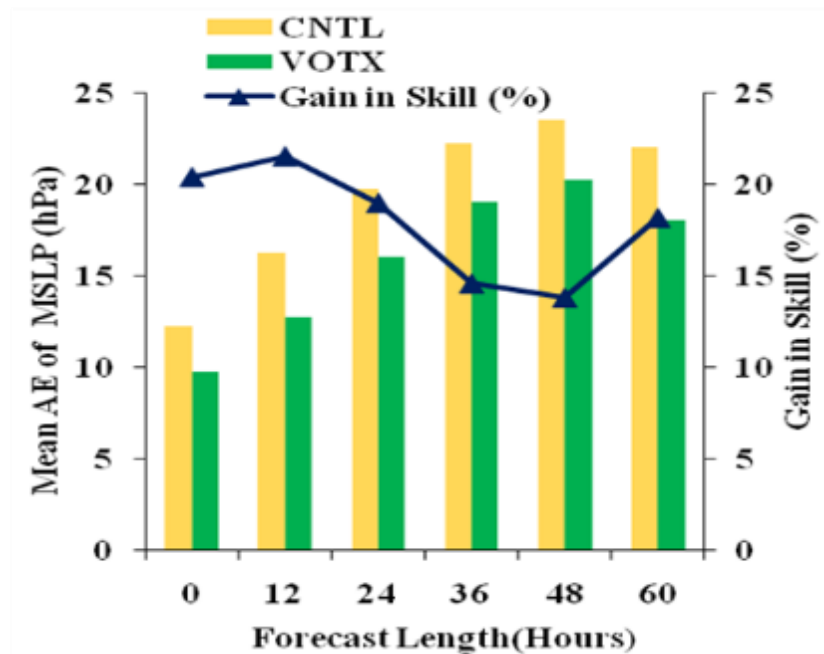


Cross Track Error (CTE)

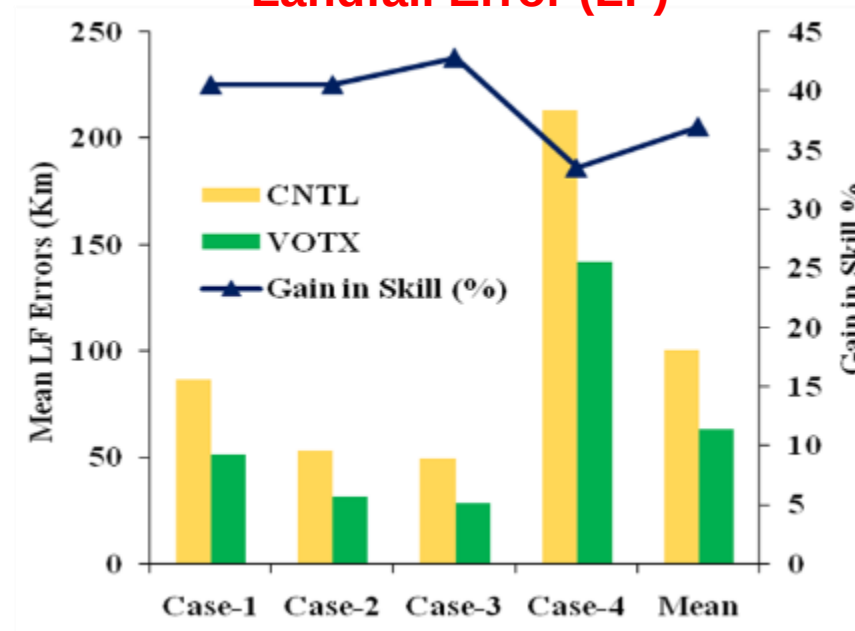


Along Track Error (ATE)



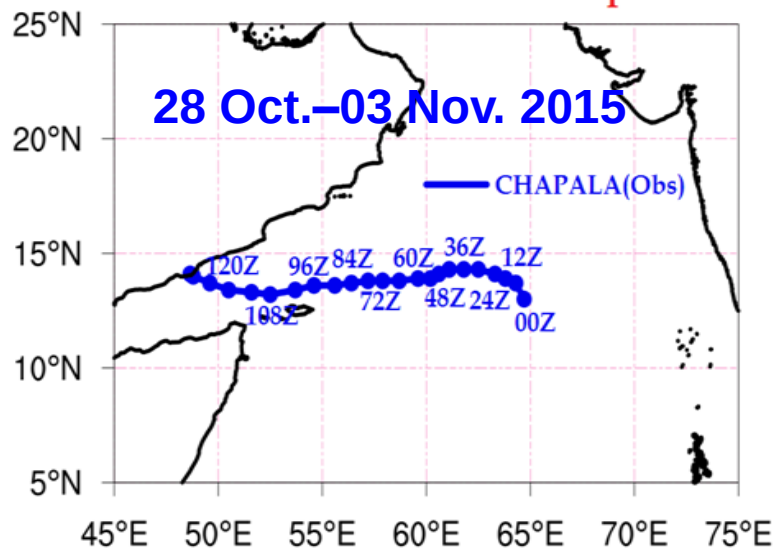


Landfall Error (LF)



Impact of VI with Satellite Data Assimilation for Predicting TC 'Chapala'

Observed track of TC 'Chapala'



Numerical Expt.

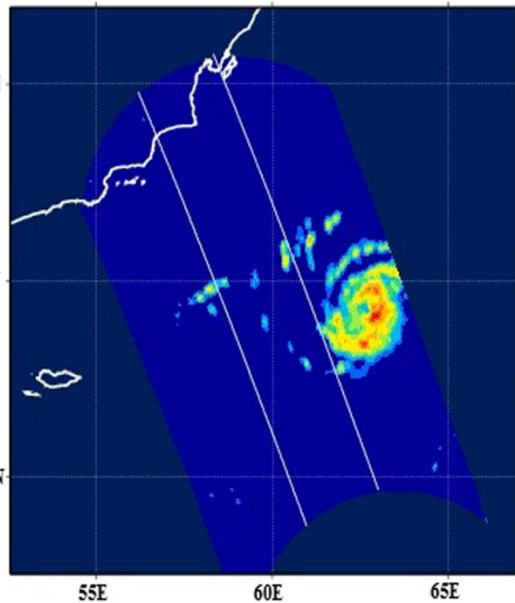
Vort-Gts: Ass. of GTS with VI

Vort-Rad: Ass. Of GTS + satellite radiance with VI

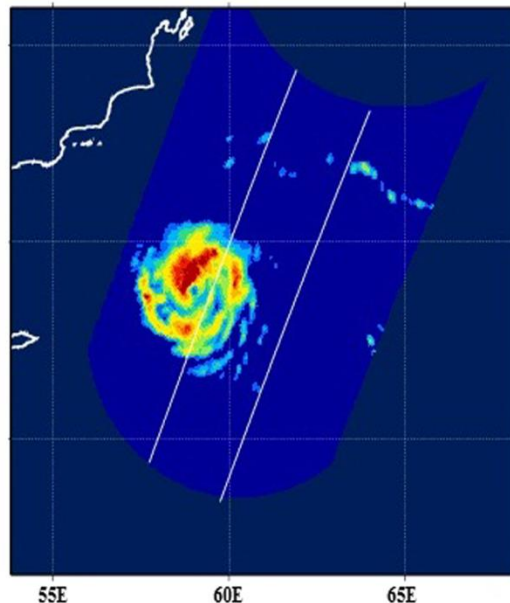
The model is run with a total of five different ICs from 29th Oct. to 03rd Nov. 2015 (24 hrs Ass. Cyclic)

GPM radar analysis

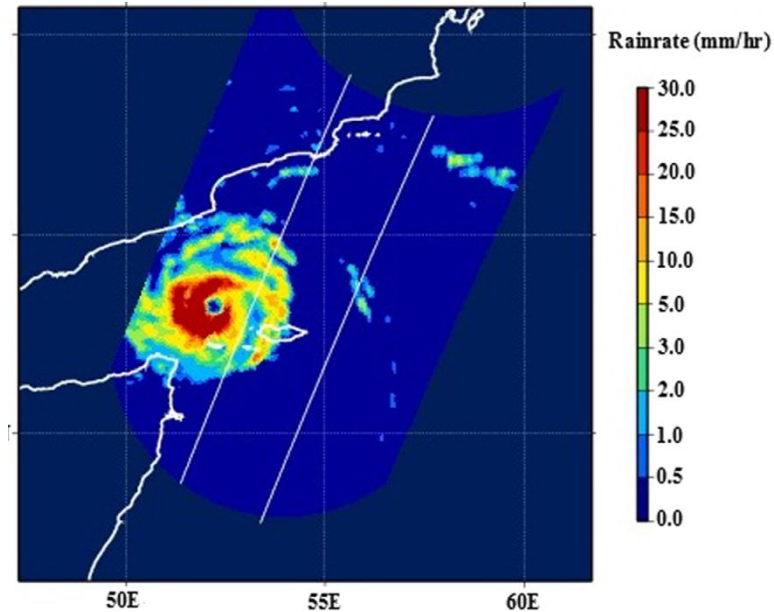
29th October



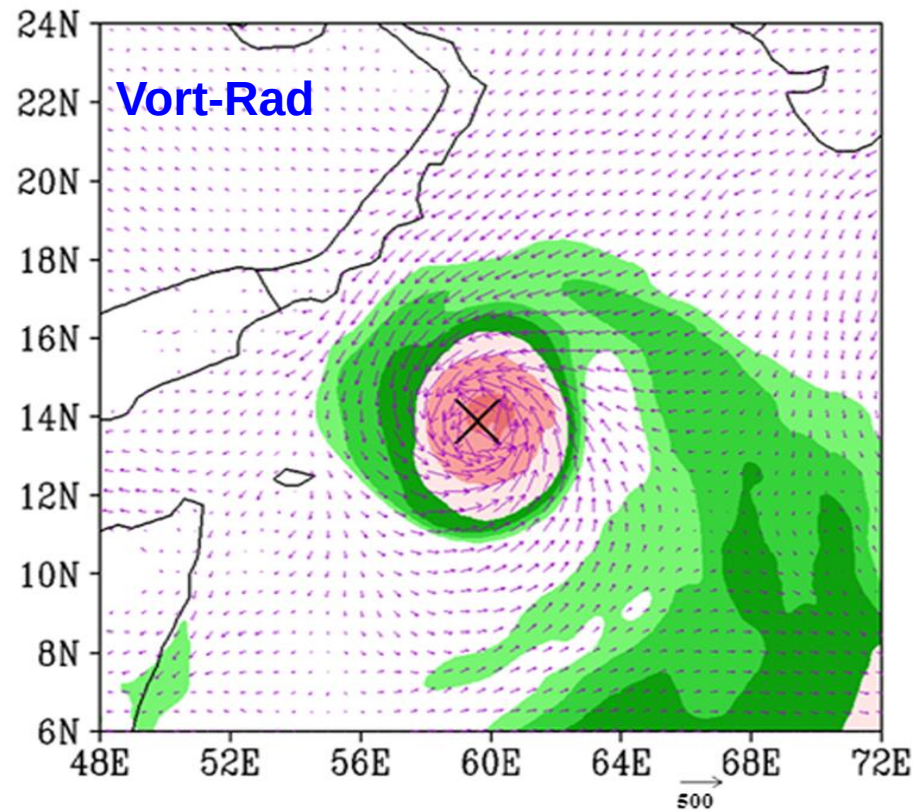
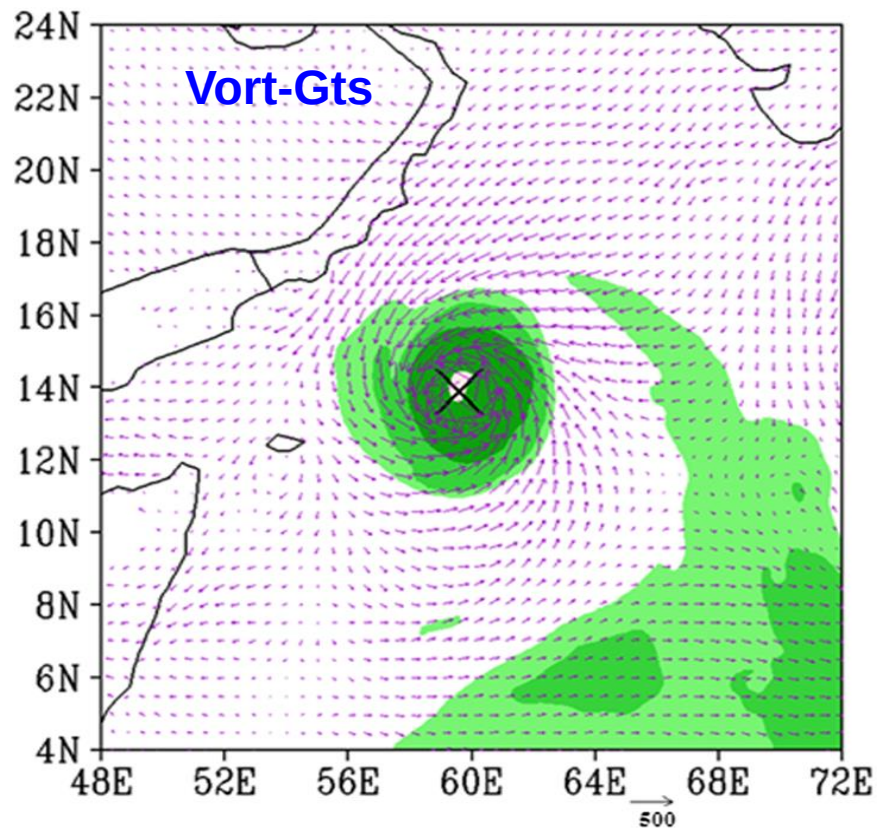
31st October



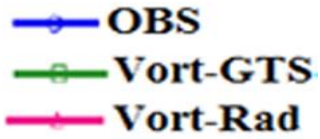
02nd November



**Moisture transport ($\text{kg m}^{-1} \text{s}^{-1}$; wind vector) and integrated total precipitable water (kg/m^2 ; shaded) valid 00 UTC 31st Oct 2015
(at Mature stage of TC)**



Tracks

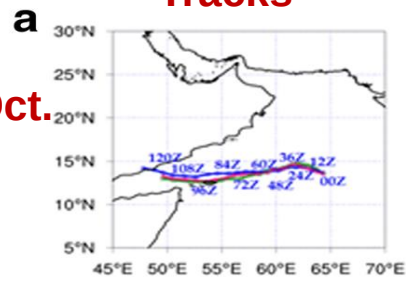


OBS

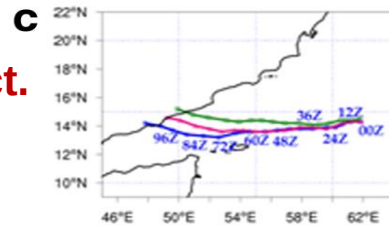
Vort-GTS

Vort-Rad

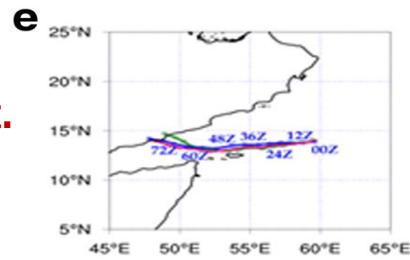
IC: 29th Oct.



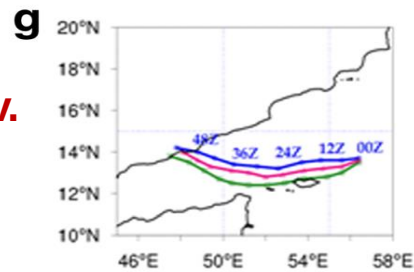
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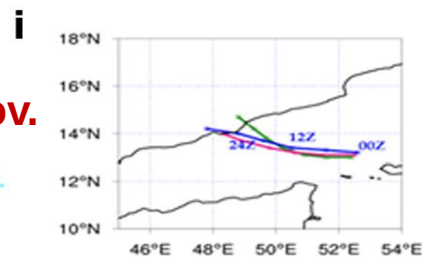
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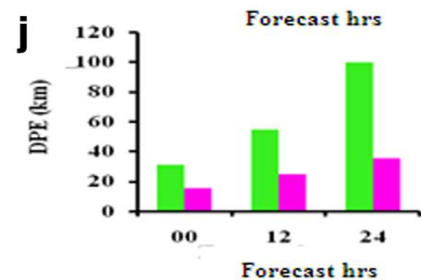
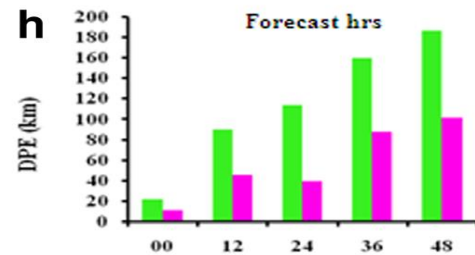
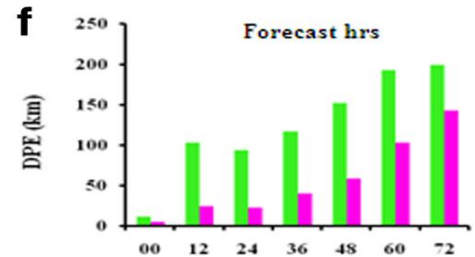
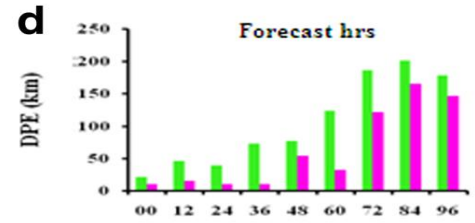
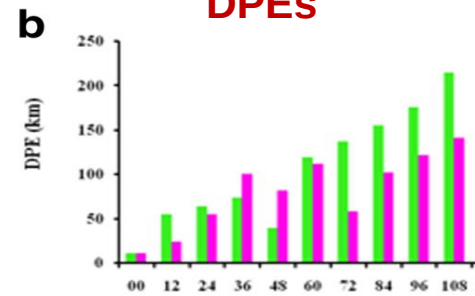
IC: 1st Nov.

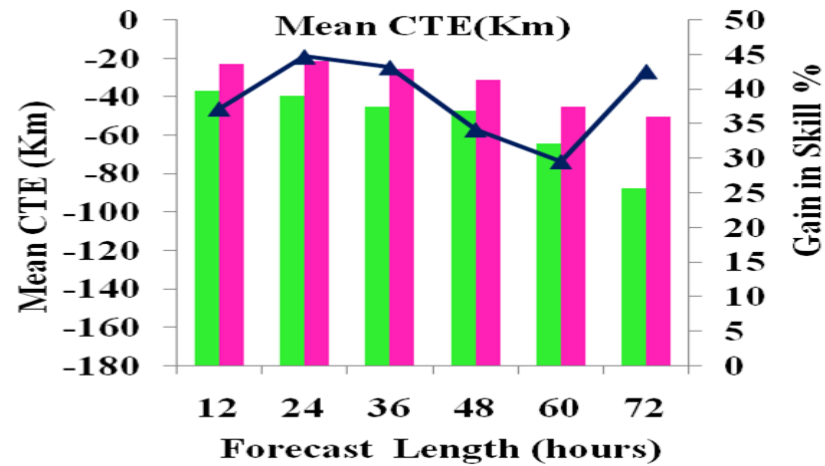
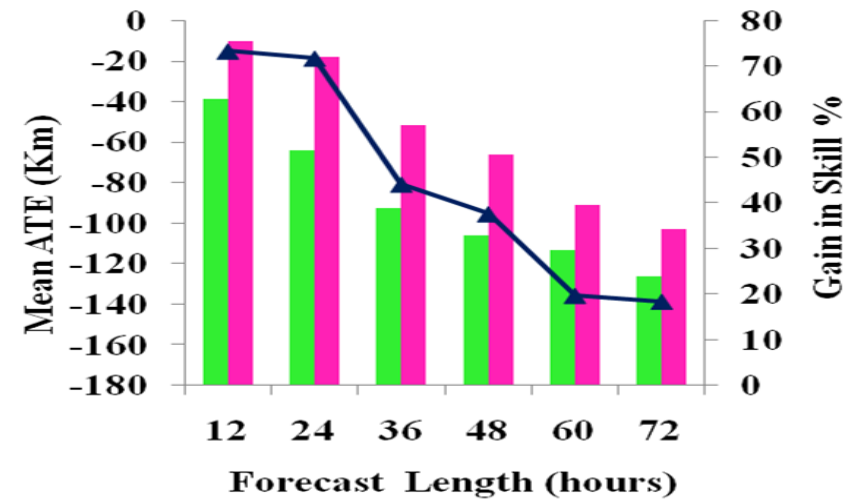
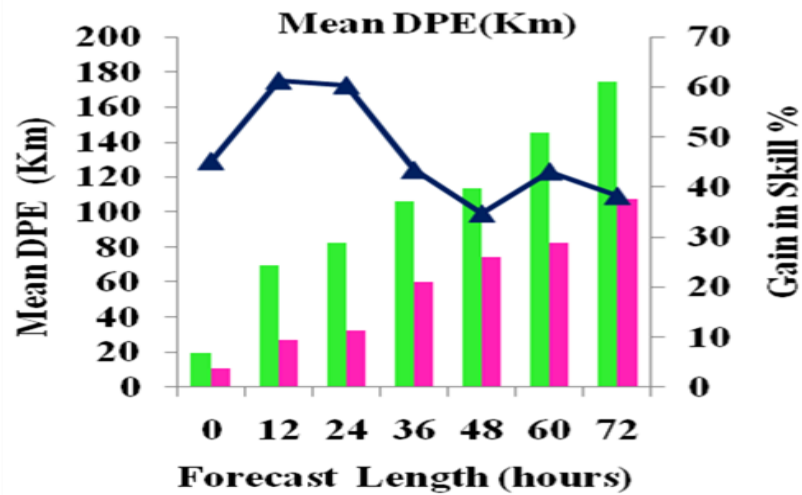


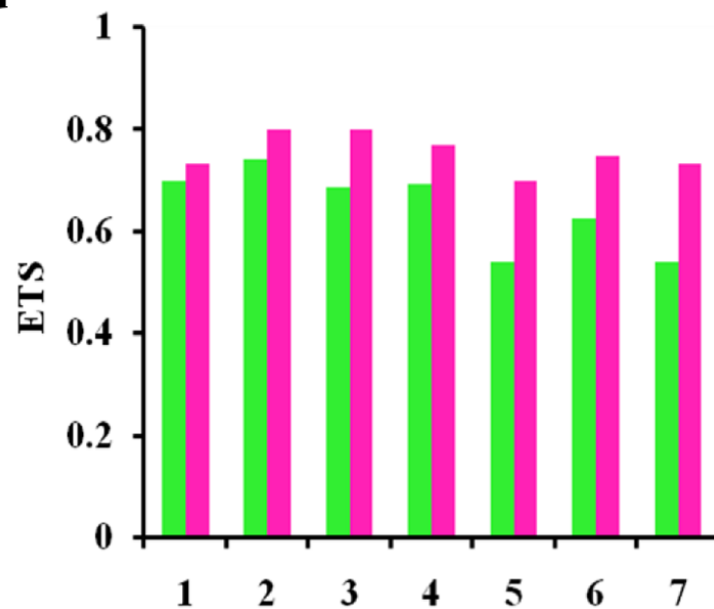
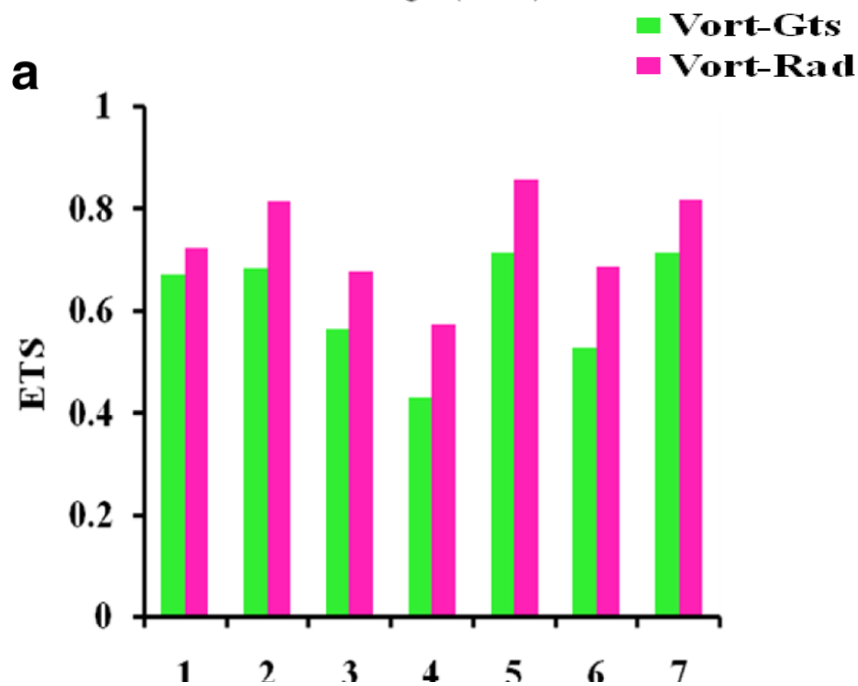
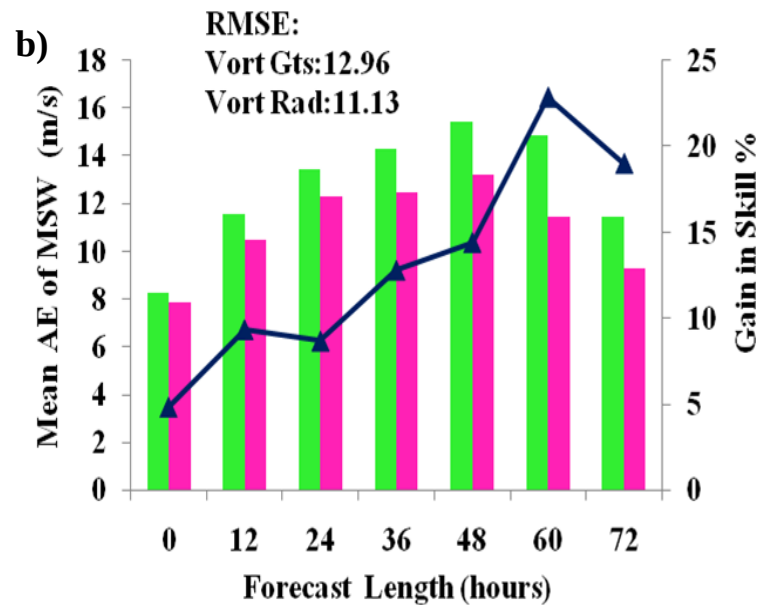
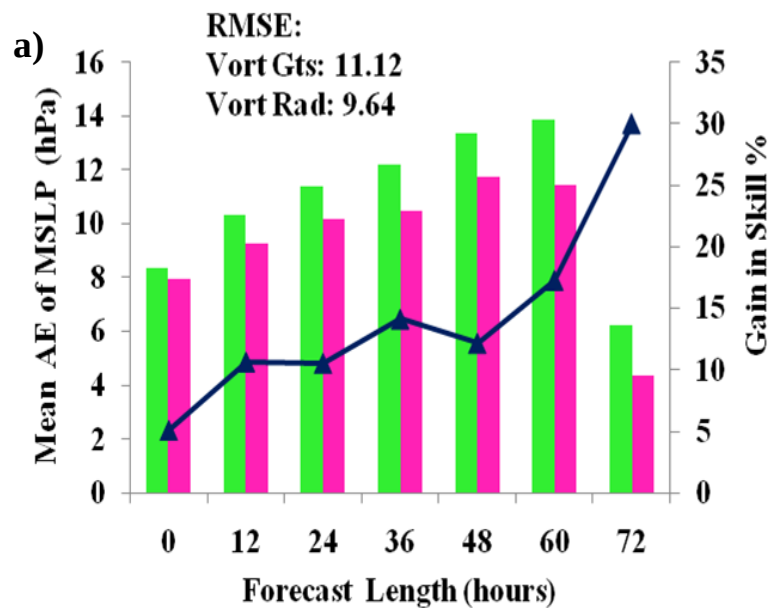
IC: 2nd Nov.

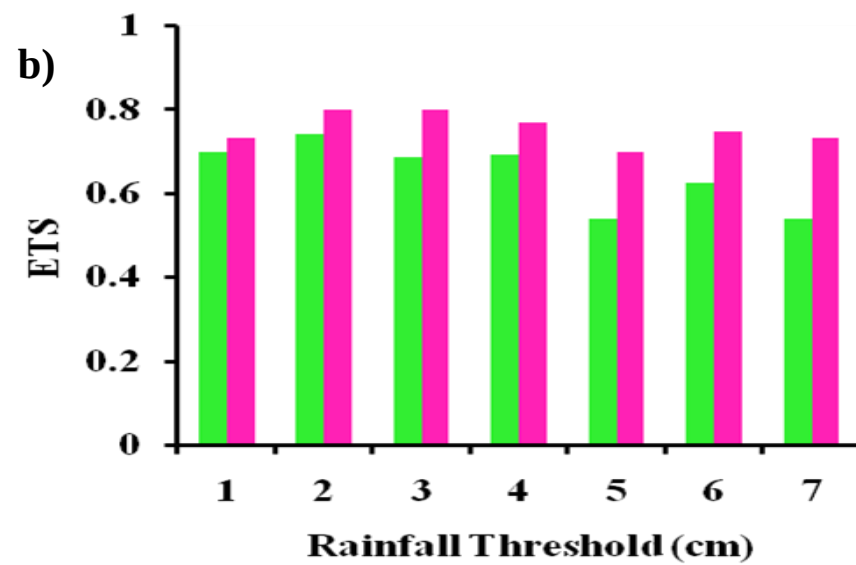
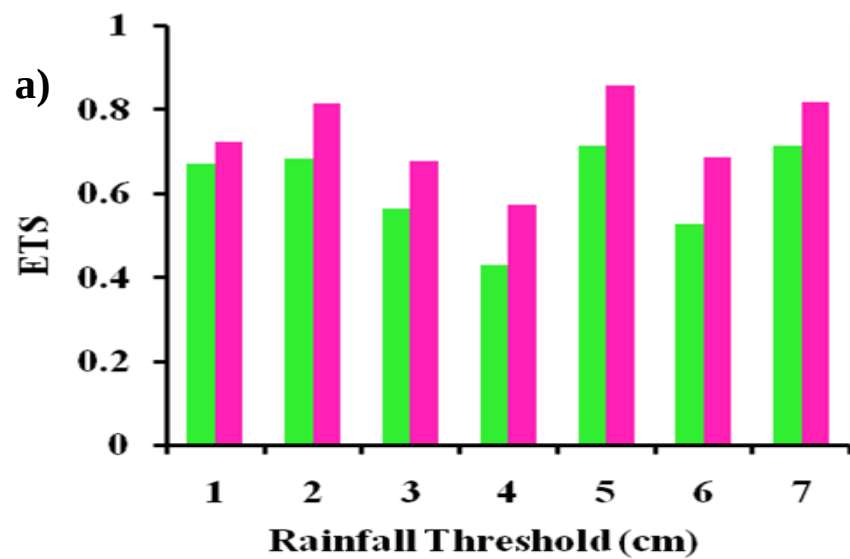


DPEs









THANKS